P84

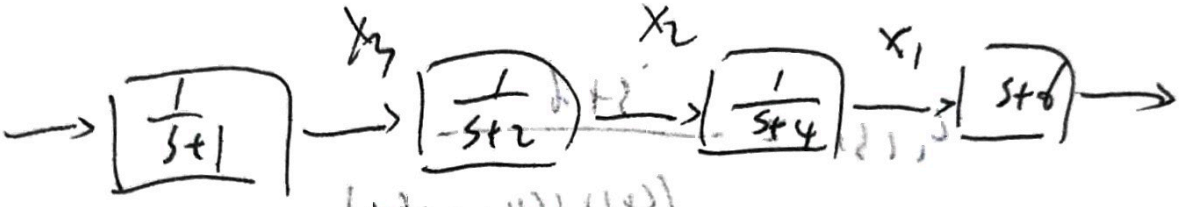
$$G(s)=\frac{s+6}{s(s+1)(s+2)(s+4)}$$

(说着用：第( )非对角)

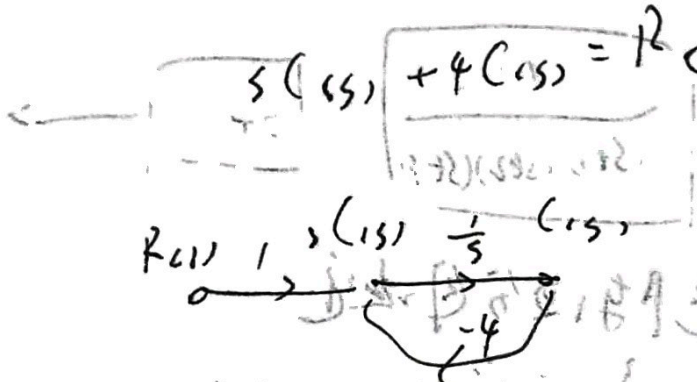
求阵

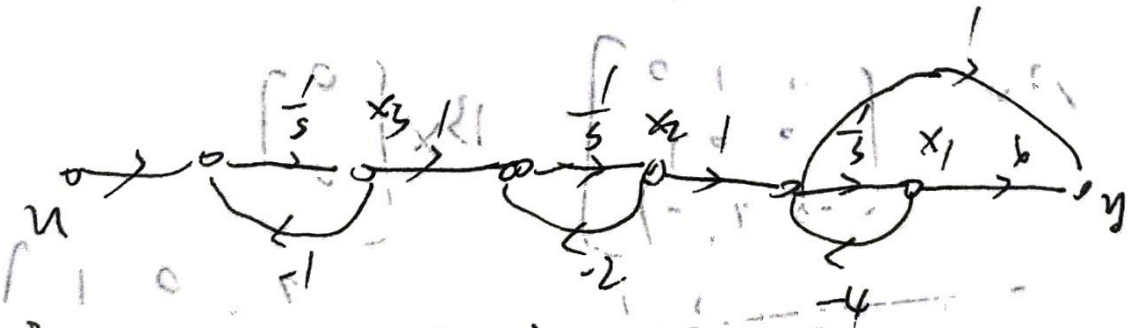
分别给出 K

比如是否一致



取一小段 $\frac{C(s)}{R(s)}=\frac{1}{s+4}$





$$\begin{pmatrix}\dot{z}_1\\ \dot{z}_2\\ \dot{z}_3\end{pmatrix}=\begin{bmatrix}-4 & -1 & 0\\ 0 & -2 & 1\\ 0 & 0 & -1\end{bmatrix}\begin{pmatrix}z_1\\ z_2\\ z_3\end{pmatrix}+\begin{bmatrix}0\\ 0\\ 1\end{bmatrix}u$$

$y=6z_1+z_2$

$=2z_1+z_2=[2\ 1\ 0]$

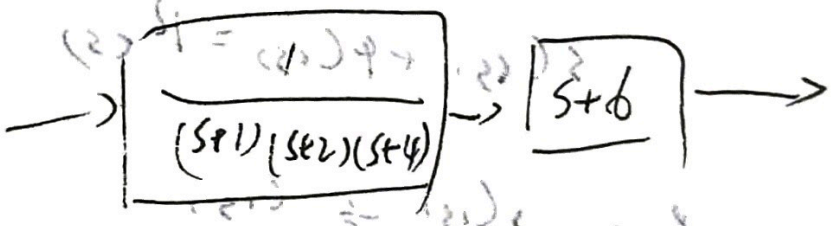
$$A_2 = \begin{bmatrix} -4 & 1 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -1 \end{bmatrix} \quad B_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$[B;\ AB;\ A^2B]$$

可控 $Q_{c2} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -3 \\ 1 & -1 & 1 \end{bmatrix}$

$$\frac{s+6}{(s+1)(s+2)(s+4)}$$

写成标准型



特征多项式 $s^3 + 7s^2 + 14s + 8$

$$A_x = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -8 & -14 & -7 \end{bmatrix} \quad B_x = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$Q_{cx} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -7 \\ 1 & -7 & 43 \end{bmatrix}$$

$$P = Q_{cz} \cdot Q_{cx}^{-1}$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -3 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 3 & 1 & -7 \\ 1 & -7 & 35 \end{bmatrix}^{-1}$$

$$\begin{bmatrix} 1 & -7 & 0 & 1 \\ -7 & 35 & 1 & -7 \\ 0 & 0 & 1 & 0 \\ 1 & -7 & 0 & -1 \end{bmatrix} \quad \begin{bmatrix} -14 & -7 & -1 \\ -7 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}^T$$

~~P~~ $\det Q_{cx} \neq 0$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -3 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 14 & 7 & 1 \\ 7 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 8 & 6 & 1 \end{bmatrix}$$

~~$K_2 = K_x \cdot P$~~

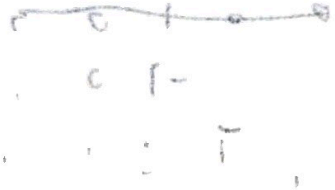
~~先求个 K_x，求问~~

先求个 K_x，然后已有 P，→ $K_2 = K_x \cdot P$ 了

求 K_2.

这个 K_2 也是操作

最下面一排



超调量 $M_p = 20.8\%$，4s 调整时间 t_s

$\zeta = 0.447$

$\omega_n = 2.236$

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

$$= -1 \pm j2$$

$$s^2 + 2s + 5 = 0$$

续 P115 草稿

引入 K. $A \to A - BK$

改变 A 矩阵，也就是系统极点

（但系统阶数不变）

只能从 $\frac{2(s+4)}{s(s+2)(s+5)} \to \frac{10(s+4)}{(s+5.4+j7.2)(s+5.4-j7.2)(s+4)}$

其中可设计一个极点与无法改变零点时消

因为零点增加超调（引入误差分量）

回看 P85 题目

$$\frac{s+6}{(s+1)(s+2)(s+4)} \to \frac{s+6}{(s^2+2s+5)(s+6)} = \frac{s+6}{(s+1+j2)(s+1-j2)(s+6)}$$

$$A_x = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -8 & -14 & -7 \end{bmatrix} \xrightarrow{\text{新的 } A_x = A_x - B_x K_x} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -8-k_{1x} & -14-k_{2x} & -7k_{3x} \end{bmatrix}$$

由新的 A_x 转为多项式

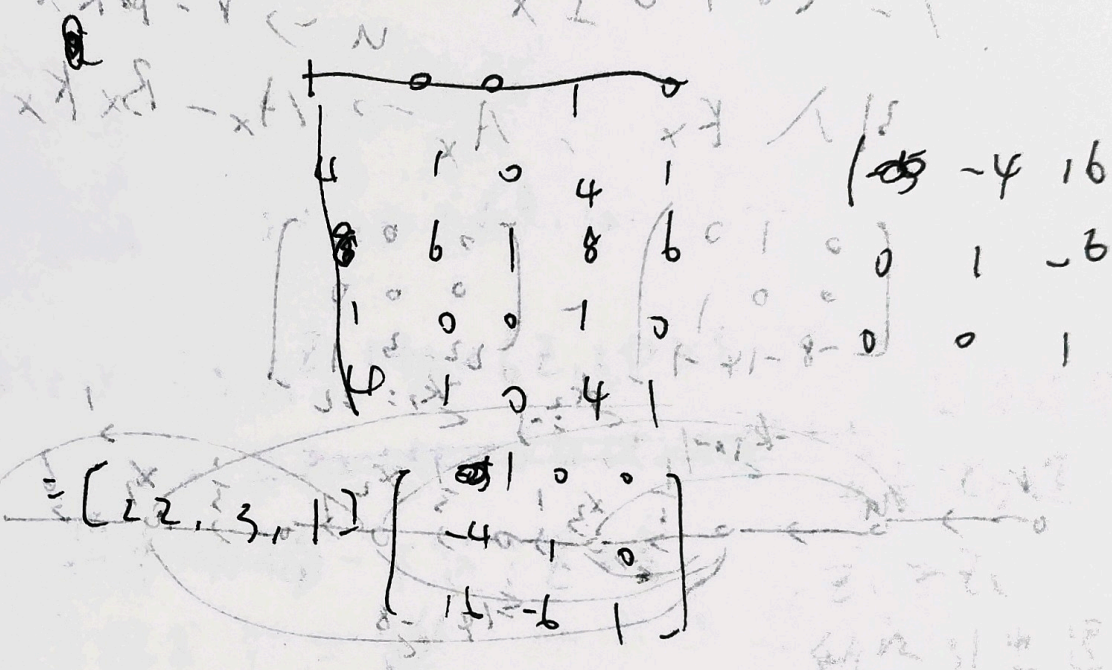
$$s^3 + (7+k_{3x})s^2 + (14+k_{2x})s + (8+k_{1x}) = 0$$

与 $(s^2+2s+5)(s+6) = s^3 + 8s^2 + 17s + 30$

$7+k_{3x}=8$ $14+k_{2x}=17$ (14) $8+k_{1x}=30$ $K_x = [22, 3, 1]$

$$K_2 = K_x \cdot P^{-1}$$

$$= [22, 3, 1] \begin{pmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 8 & 6 & 1 \end{pmatrix}^{-1}$$



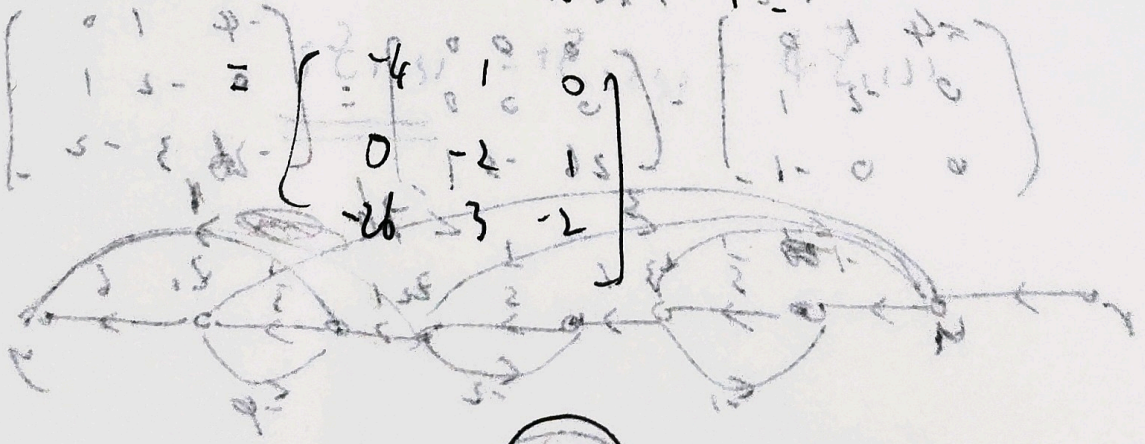
$$= [22, 3, 1] \begin{pmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 16 & -6 & 1 \end{pmatrix}$$

$$= [26, -3, 1]$$

$$A_2 \rightarrow \dot{x}_2 = B_2 K_2$$

$$\begin{pmatrix} -4 & 1 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -1 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} [26, -3, 1]$$

$$\begin{pmatrix} -4 & 1 & 0 \\ 0 & -2 & 1 \\ -26 & 3 & -2 \end{pmatrix}$$



可用矩阵型.

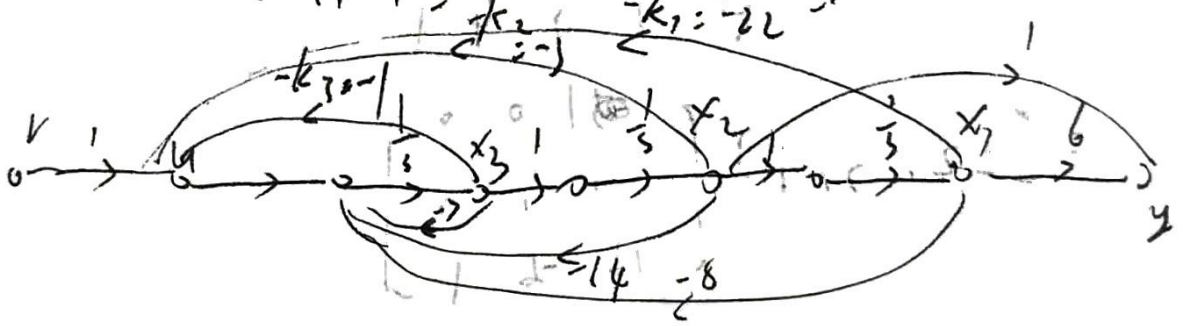
$$\dot{x} = Ax + Bu = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -8 & -14 & -7 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = [1 \ 0 \ 0]\, x$$

$u \to r - BKx$

引入 Kx, $A_x \to A_x - B_x K_x$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -8 & -14 & -7 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 22 & 3 & 1 \end{bmatrix}$$



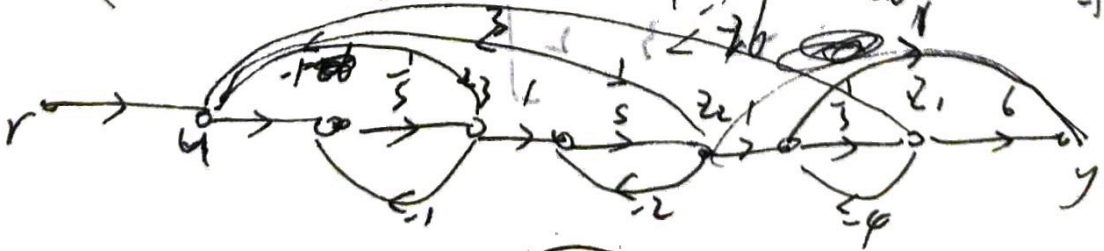
等效：非相变量

$$\dot{z} = A_z z + B_z u = \begin{bmatrix} -4 & 1 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -1 \end{bmatrix} z + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = C_z z = [1 \ 0 \ 0]\, z$$

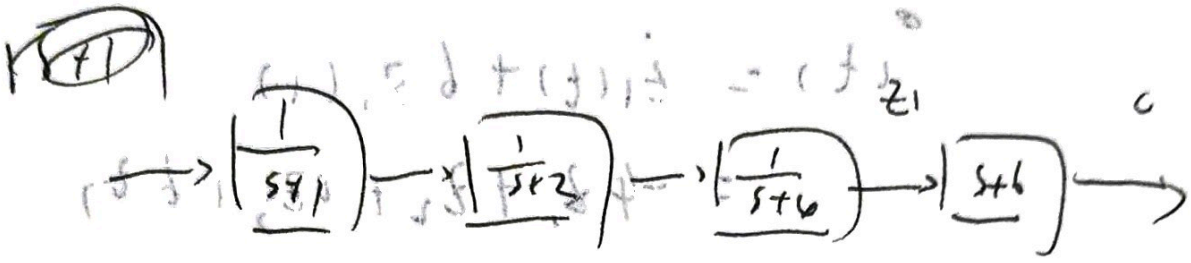
$$K_z = KP^{-1} = [26, -3, 1]$$

$$\begin{bmatrix} -4 & 1 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -1 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 26 & -3 & 1 \end{bmatrix} = \begin{bmatrix} -4 & 1 & 0 \\ 0 & -2 & 1 \\ -26 & 3 & -2 \end{bmatrix}$$



2

对于P84图4.1(b)



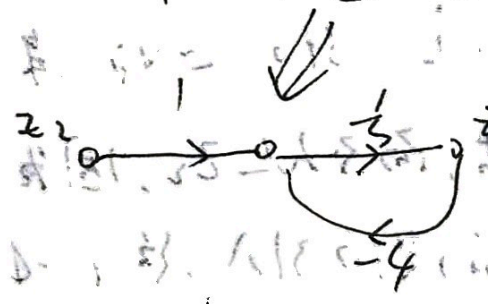
$$z_1(s)\cdot(s+6)=c(s)$$

$$sz_1(s)+6z_1(s)=c(s)$$

$$\dot{z}_1(t)+6z_1(t)=c(t)$$

~~$z_2(t)+6z_1(t)=c(t)$~~ 当写成串联并不满足

再来明确一下运算是啥



z_2 与 z_1 的关系

$$z_2(s)\cdot\frac{1}{s+4}=z_1(s)$$

$$z_2(s)=(s+4)z_1(s)$$

$$sz_1(s)=+z_2(s)-4z_1(s)$$

$$C(s) = \phi(s+b) z_1(s)$$

$$\dot{c}_1(t) = \dot{z}_1(t) + b z_1(t)$$

$$= -4 z_1 + z_2 + 0 z_3 + b z_1$$

$$= 2 z_1 + z_2 + 0 z_3$$

$$= [2, 1, 0] \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

相变量型从 z_1 或 z_1 后的 sX_1 引都可以，因为 sX_1 只有 z_1 无其他量

串联型如果按 $s+b$ 这个保留 b 最直观的是从 sz_1 引出，而不从 z_2，因为 z_2 与 sz_1 引出有除了 z_1 的其他引入，这个 $-4z_1$，

如果要从 z_1, z_2, z_3 引到 y，要自先求 C，这里 C 就不是 $[b, 1, 0]$ 而是 $[2, 1, 0]$